Introduction and Design Goal

The goal of this project was to design and implement a circuit that suppresses the G string at about 196 Hz. To accomplish this, our team designed, tested, and implemented a band-stop filter using hand calculations, simulation, breadboard testing, PCB layout, soldering, and final oscilloscope testing. Since the target frequency was 196 Hz, our design used a bandstop filter centered on that frequency. A band-stop filter was chosen because the goal was to attenuate, rather than amplify, the 196 Hz frequency component. Low- and high-pass filters are unsuitable because they would also attenuate frequencies outside the G-string range. A sharper notch allows the circuit to target the unwanted frequency more precisely, whereas a wider notch would affect a larger portion of the surrounding musical signal.

Filter Design, Component Selection, and Simulation

We selected the component values to place the center frequency of the band-stop response near 196 Hz, which is the targeted frequency of the G string. The filter used four 0.22 μF unpolarized capacitors, four 10 μF unpolarized capacitors, a 68k Ω resistor, three 1k Ω resistors, a 9k Ω resistor, two trimmers at 3.612k Ω and 1.845k Ω, LM324, and TC7660 for frequency adjustment. Five 1x2 0.1" vertical pin headers were also included to provide external connections to the PCB. Using the filter design equations, we chose resistor and capacitor values that produced a theoretical notch frequency close to the G-string fundamental. Using the notch filter equation, $f_0 = \frac{1}{2\pi RC}$ and the selected component values of C = 0.22μF and an initial resistance of approximately R = 3.612kΩ, the theoretical notch frequency was calculated to be near the target G-string frequency of 196 Hz. During testing, the trimmer resistor was further adjusted to approximately 3.7kΩ to fine-tune the notch response and better align the attenuation with the desired 196 Hz frequency. These values were verified in LTspice before the circuit was built. However, as expected, physical components did not perfectly match the ideal setup because small differences between the calculated and measured values were expected due to resistor and capacitor tolerances.

Before building the circuit physically, we simulated the filter response in LTspice. The simulation showed a clear reduction in output amplitude near the target frequency of 196 Hz. Frequencies outside the notch region were attenuated significantly less, as intended.

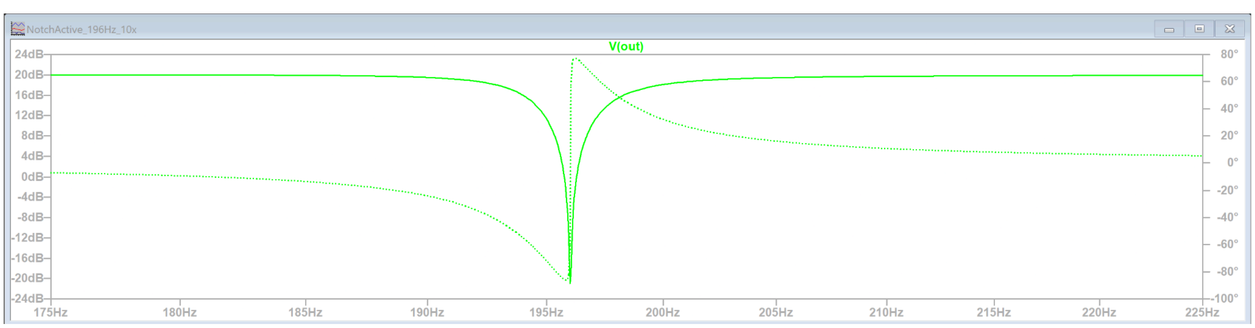

Figure 1: The LTSpice simulation shows the band-stop behavior near the target frequency of 196 Hz.

The circuit was first implemented through breadboard initial design/testing in the lab during Week 13.

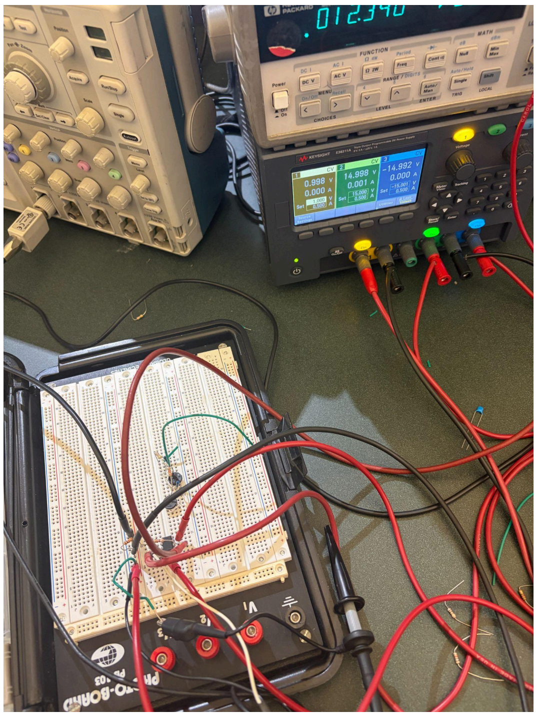

Figure 2: Gain stage implementation to amplify the input signal prior to filtering.

The circuit schematic was then created and validated in KiCad's schematic editor to ensure correct connectivity and functionality before proceeding to PCB design.

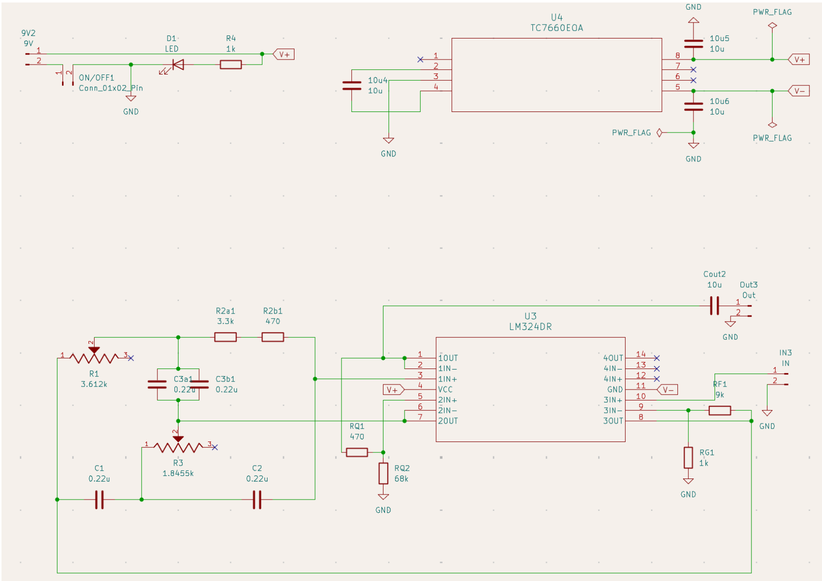

Figure 3: KiCad schematic of the active twin-T band-stop filter circuit with LM324 op-amp and feedback network.

After completing the schematic, the PCB layout was designed in KiCad by placing components and routing traces to match the intended circuit.

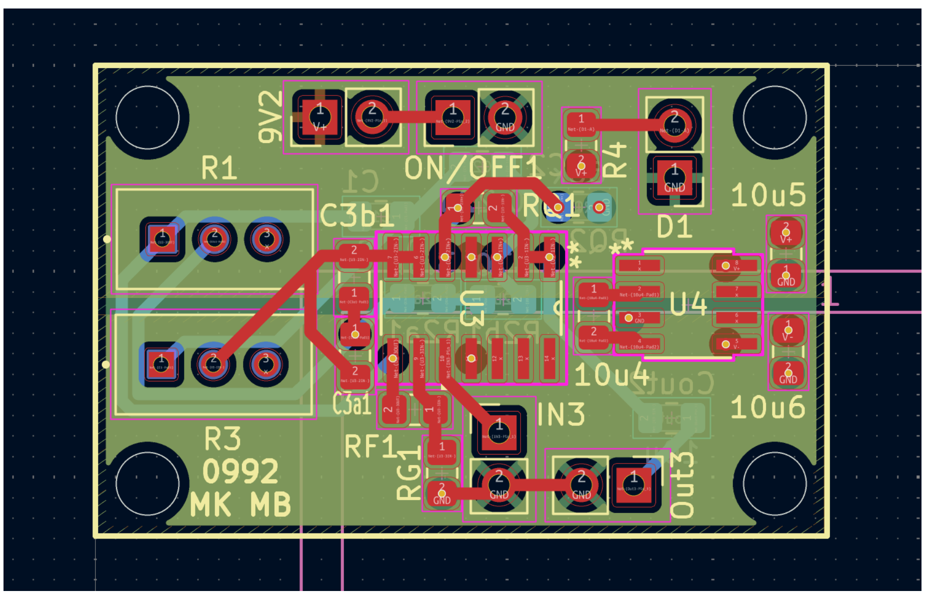

Figure 4: KiCad PCB layout showing component placement and routed traces for the active band-stop filter circuit.

PCB Physical Design, Assembly, and Guitar Integration

After validation in KiCad's schematic layout, the PCB model was used to physically match the schematic's topology to the PCB's physical dimensions and layering, enabling the PCB to be printed later. The PCB's layout organized the components and traces so the circuit could be fabricated, soldered and placed inside the guitar system.

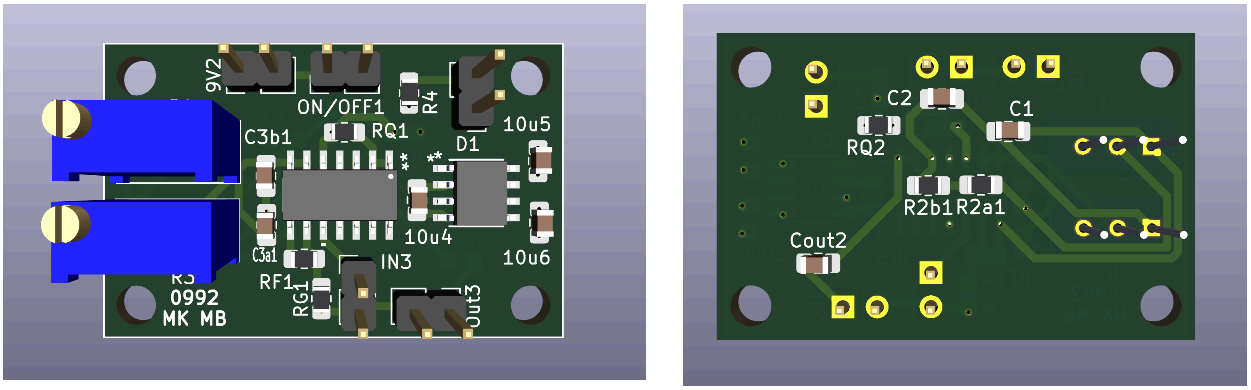

Figure 5: The digital 3D PCB model shows the component placement and board layout of the front and back before fabrication.

Once the PCB was fabricated, the circuit components were soldered onto the board and inspected. After assembly, the PCB was tested to confirm that it achieves the intended band-stop behavior.

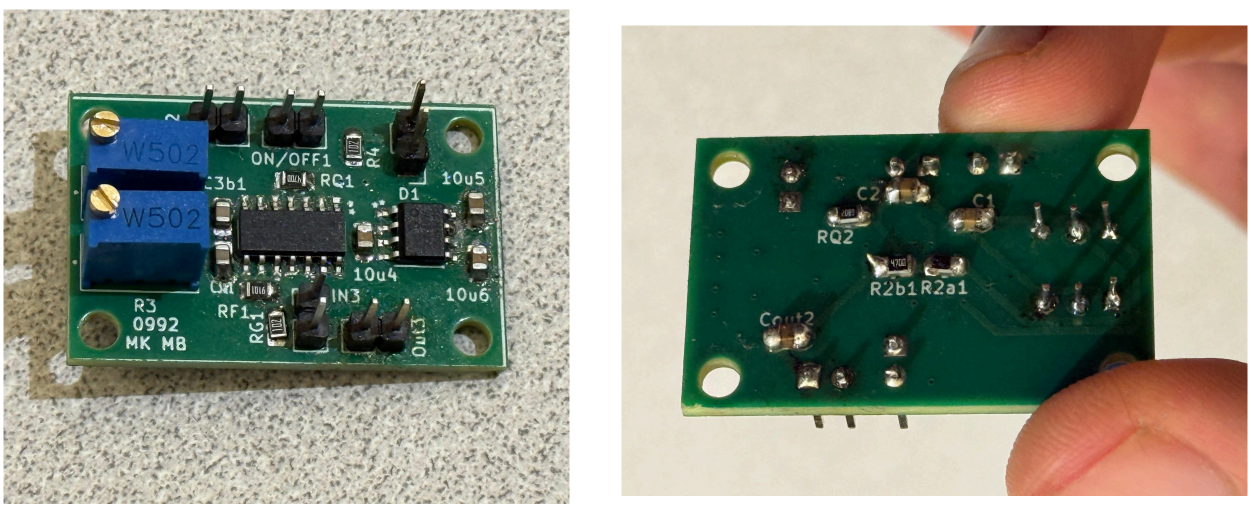

Figure 6: This figure shows the PCB components soldered onto the board.

The physical guitar assembly included the neck, strings, pickup, PCB, and speaker connection. The pickup signal was routed through the filter circuit before reaching the output, allowing the PCB to directly affect the electrical guitar signal. This connected the circuit design to the final guitar system.

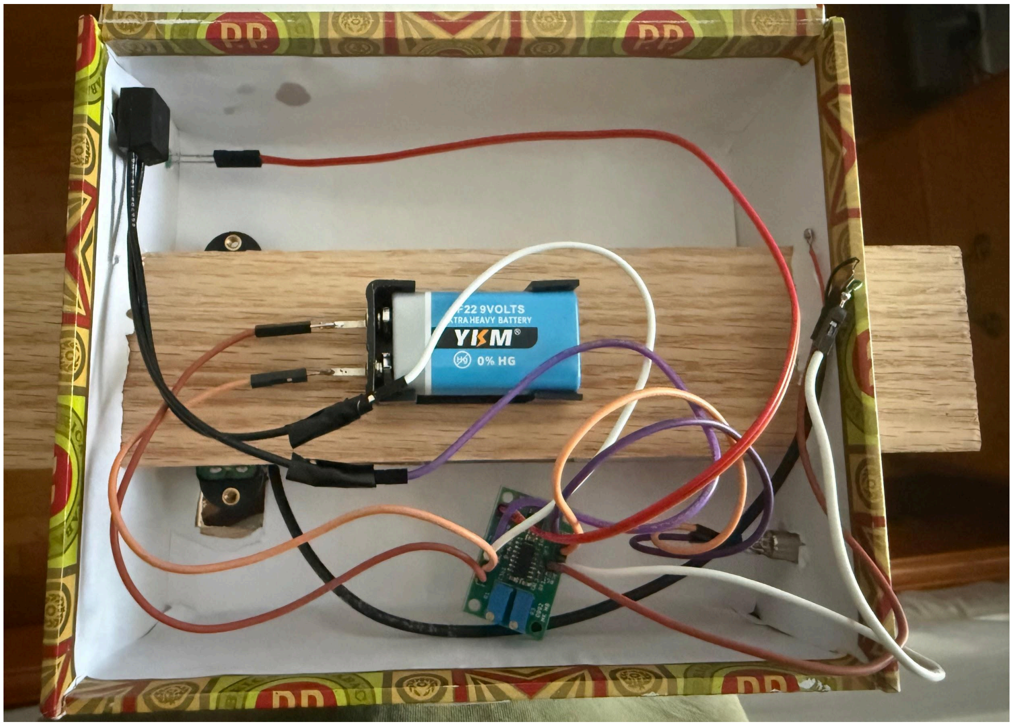

Figure 7: This figure shows the interior of the guitar with the PCB and electrical connections.

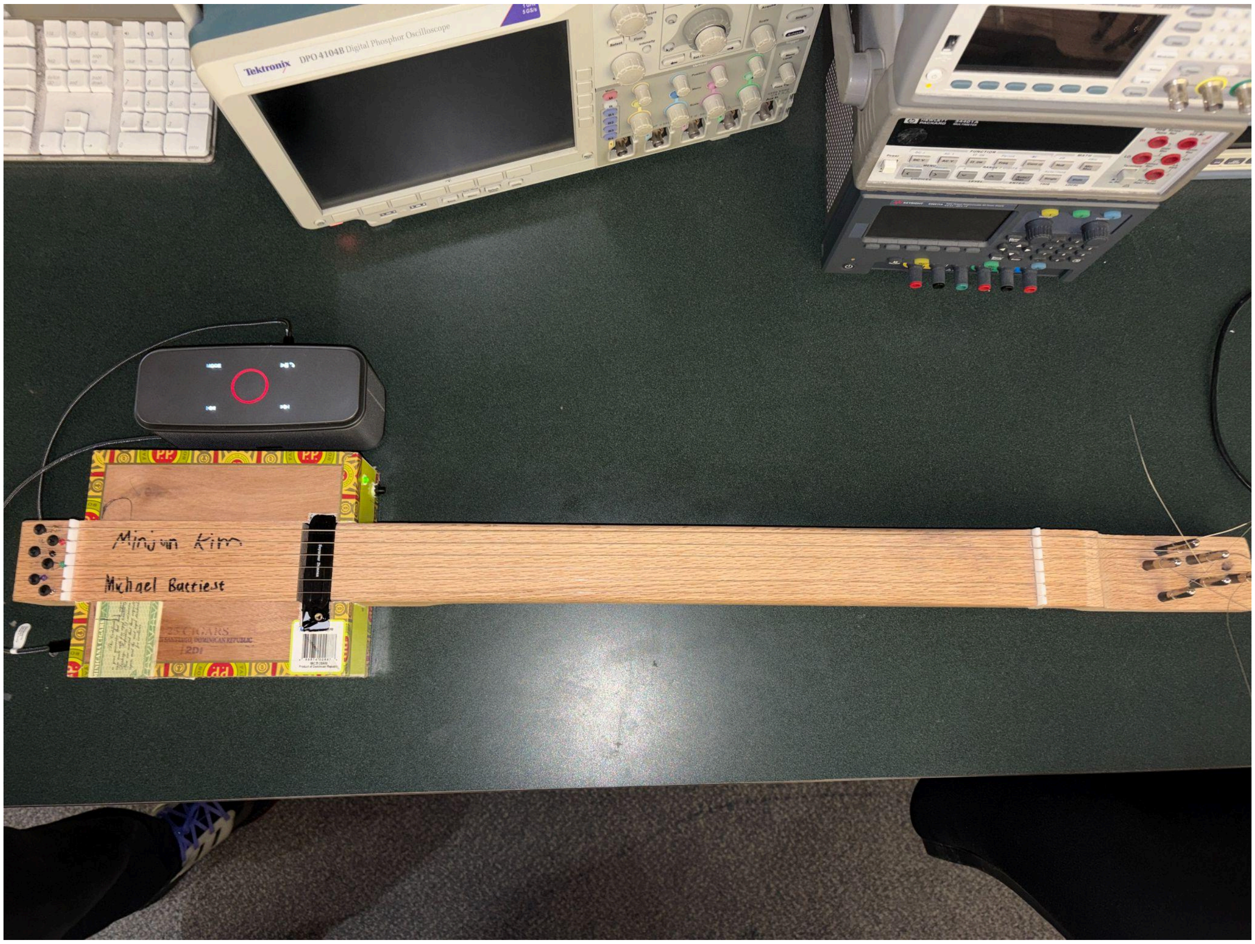

Figure 8: Final assembly of the guitar.

Link to the video of the guitar working:
https://drive.google.com/file/d/17IHlwNuX7_5Mukt3s1WqXCP0-reA4ZSN/view

Final Testing Results

In the final test, the circuit showed the expected band-stop behavior around the target frequency of 196 Hz. At 186 Hz, the output signal measured approximately 4.14 Vpp, and at 206 Hz, the output was about 3.92 Vpp. However, at 196 Hz, the output dropped significantly to approximately 0.64 Vpp. The relatively high output amplitudes were expected because the signal passed through an op-amp gain stage before filtering.

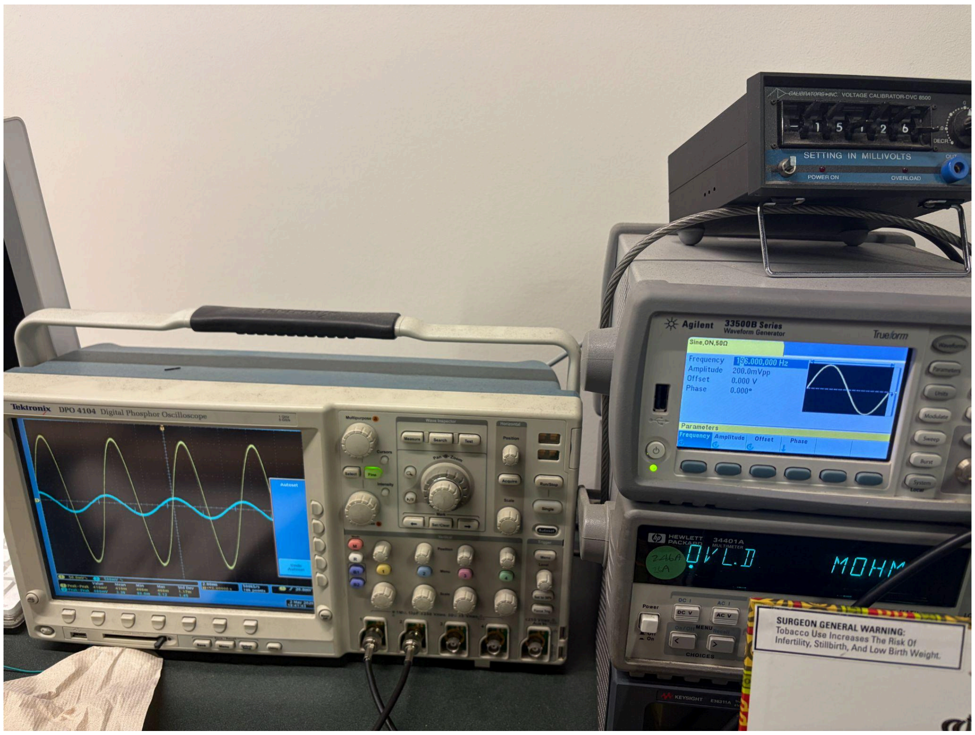

Figure 9: The oscilloscope measurement at 196 Hz shows that the output signal is significantly attenuated.

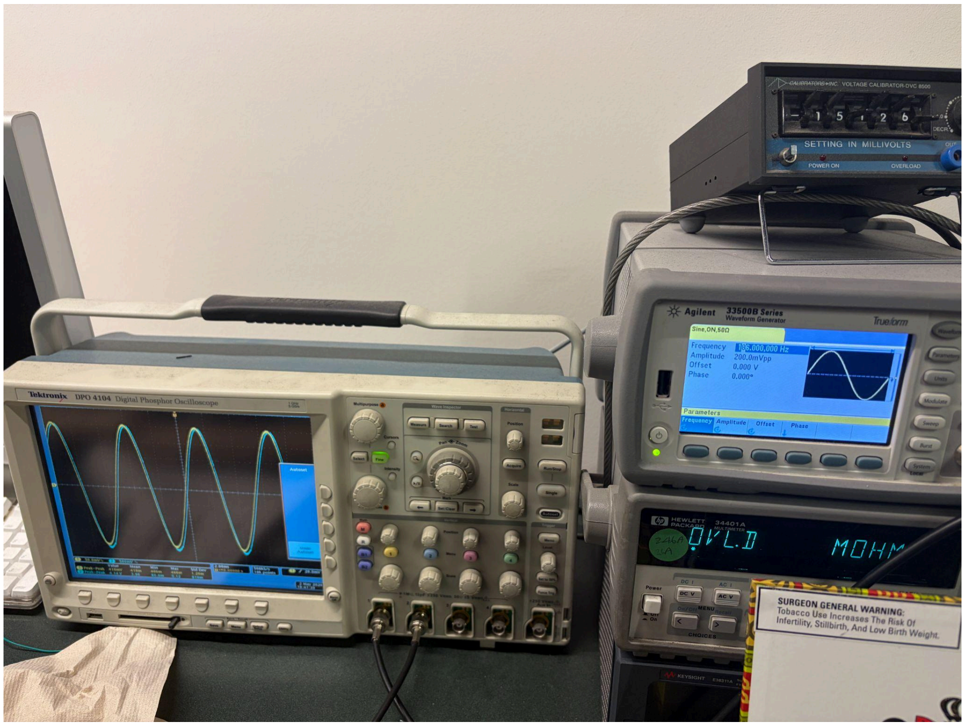

Figure 10: Oscilloscope measurement at 186 Hz demonstrating that frequencies outside the notch region are largely preserved.

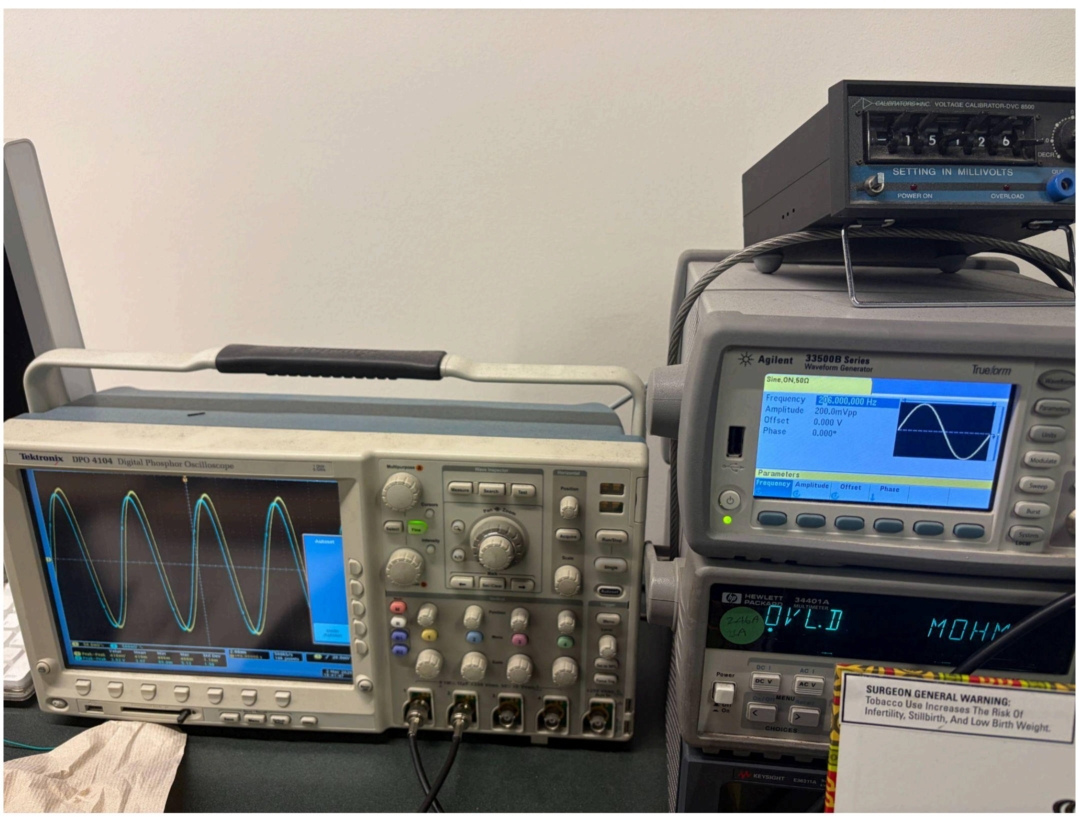

Figure 11: Oscilloscope measurement at 206 Hz demonstrating that frequencies outside the notch region are largely preserved.

This demonstrates that the circuit achieved noticeable attenuation at the target frequency compared to nearby frequencies. When the input frequency was moved away from the notch, the output remained relatively high, indicating that the filter preserved signals outside the targeted range. These results confirm that the circuit primarily suppressed the G-string frequency rather than attenuating the entire signal.

One issue during debugging was making sure the components were fully soldered to the PCB. We realized that some components were not soldered as intended during the testing phase, because we observed noise and behaviors we did not expect in the measured output on the oscilloscope. Due to our limited soldering experience, we found it difficult to properly hand-solder the small capacitors and resistors, which resulted in incomplete attachment. We had to desolder a couple of capacitors and solder them again to achieve the final outcome. After soldering, the circuit was inspected and tested with the oscilloscope to verify that the output dropped near 196 Hz.

Link to the video of the demonstration of the notch filter working:
https://drive.google.com/file/d/1LjKbJNRd8FekyDO8iNKw_yfxd0QW2MO8/view?usp=sharing

Conclusion

The project successfully demonstrated the design, simulation, PCB implementation, and testing of an active band-stop filter intended to attenuate the 196 Hz fundamental frequency of the G string. Based on the oscilloscope measurements, the circuit showed clear notch-filter behavior near the target frequency. At 186 Hz, the output was 4.14 Vpp, and at 206 Hz, the output was 3.92 Vpp, while at 196 Hz, the output dropped to 0.64 Vpp. This confirms that the filter was centered near the desired frequency and significantly attenuated signals at 196 Hz compared to nearby frequencies.

However, the filter didn't fully remove the G string, as the output at 196 Hz was lower than the surrounding frequencies and did not reach zero. This means the filter reduced the signal, but not entirely. One possible improvement would be to increase the notch depth by further tuning the filter to produce sharper attenuation at 196 Hz.

Overall, the project achieved its goal: the 196 Hz signal was attenuated, and the band-stop filter design demonstrated its effectiveness. To eliminate the G string, the next step is to push for a sharper, deeper notch at that frequency. The PCB fabrication and shipment delay limited the amount of time available to refine the design. By the time the board arrived, we did not have enough time left to iterate and dial things in properly. Given more time to debug and fine-tune, the circuit could achieve a much deeper, cleaner notch at 196 Hz.